AWS CLOUD CLUB VJIT | PARTICIPANT HANDBOOK

Cloud-Tycoon

The 5-Year Turnaround

You're walking into a burning building. A startup called PulseStream is hemorrhaging money on AWS — spending more than they earn. The previous team panicked and bailed. Now it's your turn.

Over five rounds (each = one fictional year), you and your team will make real cloud engineering decisions to drag this company from the brink of bankruptcy to global domination.

This handbook is your survival kit. Read your role section carefully. Refer to it during the game. Win because you understood the business, not because you guessed.

CTO	CFO	PM
Cloud Architect	Financial Analyst	Growth Lead
The technical mind	The bean counter	The ambitious one

FOR 2nd Year B.Tech | **VENUE** VJIT Hyderabad

HOW THIS HANDBOOK WORKS

READ THIS FIRST

This is not a cheat sheet

The answers still require **logical thinking** and applying concepts to PulseStream's specific situation. Memorizing won't win this game — reasoning will.

1. Read **Part 1 (Game Overview)** and **Part 2 (AWS Fundamentals)** — everyone needs these.
2. Read **your role's section** deeply — CTO (Part 3), CFO (Part 4), or PM (Part 5).

3. Skim the other two role sections — understanding basics helps collaboration.
4. Bookmark **Part 6 (Quick Reference)** — keep it open during the game.

PART 01

Game Overview

The PulseStream story, the 5 years, and how you win.

It's 2 AM. The CEO of **PulseStream** — a video analytics startup — just messaged you: *"We need help."* The previous team bailed. The cloud bill is spiraling. Customers are complaining. You have 5 years to fix it.

MONTHLY REVENUE	MONTHLY AWS BILL	NET POSITION	RUNWAY
$12,000	$18,000	-$6,000	10 months

The 5-year roadmap

Year / Round	Challenge
Year 1 — The Cleanup	Cut AWS waste without breaking the product. Stop the cash bleed.
Year 2 — Commitment Trap	Decide how much to commit to Reserved Instances vs staying flexible.
Year 3 — Chaotic Virality	Handle a 10x traffic spike from a viral event. Scale smart, fast.
Year 4 — Cascade Failure	Survive simultaneous AWS outage + security breach. Crisis management.
Year 5 — Strategic Warfare	Go global, compete with well-funded rivals, prepare for IPO.

WHY THIS MATTERS

Your role is your superpower

Every question is designed for a specific role — CTO, CFO, or PM. You won't be tested on the other roles' deep knowledge, but understanding the basics helps your team collaborate. **The best teams argue across roles.**

Scoring: Cloud Valuation Index (CVI)

Metric	Weight	What it measures
Net Cumulative Profit	50%	Total revenue minus total AWS costs across all rounds
Uptime Percentage	30%	How well your architecture survived without losing revenue
Cost Efficiency	20%	Revenue generated per dollar of AWS spend

PART 02

AWS Fundamentals

Concepts every team member should know — CTO, CFO, and PM.

How AWS charges you

AWS does not charge a flat monthly fee. **You pay for what you use, when you use it.**

Billing dimension	Examples	Key insight
Compute (EC2, Lambda)	Per hour or per second depending on type	Unused instances still cost money
Storage (S3, EBS)	Per GB stored per month	Unattached volumes are pure waste
Data Transfer Out	Per GB sent from AWS to internet	Video delivery costs scale with views
Managed Services	Per request/hour/unit (RDS, ElastiCache)	Often cheaper than managing yourself
Networking	Cross-AZ data transfer charged per GB	Services spread across AZs = hidden cost

EC2 purchase models — the most important decision

Model	Discount	Best used for
On-Demand	0% (full price)	Unpredictable workloads; new services; testing
Reserved Instance (1-yr)	~40% off	Stable, predictable workloads running 24/7
Reserved Instance (3-yr)	~60% off	Long-term stable workloads — high commitment risk
Savings Plans	~20-40% off	Flexible: applies across EC2, Lambda, Fargate
Spot Instances	~70-90% off	Fault-tolerant batch jobs — can be interrupted 2 min notice

RULE

RI Break-Even threshold

RIs save money **only if the instance runs at least 55-60% of the time.** Below that, On-Demand is cheaper. Dev/staging environments that shut down nights and weekends rarely hit this.

PRO TIP

Standard vs Convertible RIs

Standard RIs: cheapest, locked to one instance type and region. Can be sold on RI Marketplace.
Convertible RIs: slightly less discount, but can exchange for different instance family. Use when unsure about your instance type.

The 5 AWS services you must know

Service	What it does
EC2	Virtual servers. The main compute workhorse. Choose size, pay per hour.
S3 (Simple Storage)	Object storage for files, videos, backups. Extremely cheap, scales infinitely.
RDS (Relational DB)	Managed SQL database. AWS handles backups, patches, and failover.
Auto Scaling	Automatically adds/removes EC2 instances based on CPU, traffic, or custom metrics.
CloudFront	CDN. Caches content at 400+ global edge locations. Reduces latency and origin load.

Key AWS cost management tools

Tool	Purpose
AWS Cost Explorer	Visual dashboard of spend by service, region, time. First stop for cost analysis.
AWS Budgets	Set spending alerts that email you when costs approach a threshold.
Cost Allocation Tags	Tag resources by team/feature/environment to track what each area costs.
Trusted Advisor	Automated recommendations for cost, security, reliability. Flags underutilized EC2.
Cost & Usage Report	Most granular billing data — every resource, every hour. For deep analysis.
Compute Optimizer	Analyses metrics and recommends right-sized instance types before you commit.
Cost Anomaly Detection	ML-powered tool that detects unusual spend spikes automatically. Free to use.

Reliability: AZ, Region, Multi-AZ, Multi-Region

Understanding AWS geography is **critical for Year 4** (Cascade Failure).

Concept	What it means
Availability Zone (AZ)	One physical data centre within a region. AWS has 2-6 AZs per region (e.g., us-east-1a, 1b, 1c).
Region	Geographic area containing multiple AZs. E.g., us-east-1 (Virginia), eu-west-1 (Ireland).
Multi-AZ	Running resources across 2+ AZs. Protects against one data centre failing. Standard for production.

Multi-Region	Running resources in 2+ regions. Protects against entire region down. Expensive but critical for 99.99%.
RTO (Recovery Time)	How long can PulseStream be down before serious damage? Defined in hours or minutes.
RPO (Recovery Point)	How much data can be lost in a failure? E.g., RPO=15 min means restore up to 15 min before failure.

RULE

Uptime SLA cheat sheet

99.9% = 43.8 min downtime/month | **99.95%** = 21.9 min/month | **99.99%** = 4.4 min/month A 4-hour outage violates 99.9% SLA — this matters for Year 4.

PART 03 · CTO ROLE

The Cloud Architect

You design the architecture, choose AWS services, and make technical trade-offs. You answer: "How do we build it?"

YEAR 1 · Right-Sizing and Cleanup · CTO

Right-sizing means choosing an instance type that matches your actual workload. Running **m5.xlarge at 12% CPU** is like hiring a truck driver to deliver a pizza — wasteful.

Instance family	Typical use case
t3/t4g (Burstable)	Low-to-moderate workloads with occasional spikes. Good for dev, staging.
m5/m6 (General Purpose)	Balanced CPU/memory. Most production web and app servers.
c5/c6 (Compute Optimised)	CPU-intensive: video encoding, high-traffic APIs.
r5/r6 (Memory Optimised)	Large in-memory databases, caching layers, analytics.

CLEANUP ORDER

The 6-step cleanup process

1. Run AWS Cost Explorer to find biggest cost services
2. Check Trusted Advisor for underutilised instances
3. Right-size EC2 (move from m5.xlarge to t3.medium)
4. Delete unused EIPs, unattached EBS volumes, idle snapshots
5. Schedule dev/staging shutdowns with AWS Instance Scheduler
6. Migrate EBS gp2 volumes to gp3 (same performance, ~20% cheaper)

PRO TIP

Dev/Staging cost rule

Dev and staging environments are typically idle 65-70% of the time (nights, weekends). Scheduling shutdowns during off-hours with AWS Instance Scheduler **reduces cost by 60-70%** with zero impact on development.

YEAR 2 · Reserved Instances Strategy · CTO

RI Type	Best for	Risk
1-year Standard	Stable, proven workloads you're sure won't change	Locked to instance type and region for 12 months
1-year Convertible	Workloads that may evolve — can exchange for different family	Slightly smaller discount than Standard
3-year Standard	Long-term stable baseline (e.g., main database)	Very high risk for startups — 3-year bets are risky
Savings Plan	Mixed workloads across EC2 + Lambda + Fargate	Less specific than RI — may not fully cover all types

RULE

The Mixed Fleet Strategy (AWS best practice)

70% Reserved Instances (predictable baseline)
20% On-Demand (variable headroom)
10% Spot Instances (fault-tolerant batch jobs)

This is the AWS recommended best practice and appears in Year 2 questions.

PRO TIP

RI utilisation rate

If your RI utilisation in Cost Explorer is **below 60%**, you over-committed. Goal is 80%+. Below 55%: On-Demand would have been cheaper.

YEAR 3 · Auto Scaling and Traffic Spikes · CTO

A 10x traffic spike in 20 minutes **cannot be handled manually.** The architecture must respond automatically.

Component	How it helps during a spike
Auto Scaling Group	Launches new EC2 instances automatically when CPU > 70% for 2 consecutive min.

Application Load Balancer	Distributes incoming requests across all available instances. Scales automatically.
Amazon SQS Queue	Absorbs upload spikes by queuing jobs. Workers process at own pace. No dropped requests.
RDS Read Replicas	Take read traffic off primary database with zero downtime. Add in minutes.
ElastiCache (Redis)	Cache frequent queries in memory (microsecond response). Massively reduces DB load.
CloudFront	Serves cached video from edge locations. Reduces origin load by 80%+ with good cache hit rate.

RULE

Spike Response Priority Order

1. Auto Scaling handles compute (automatically if configured)
2. SQS queue absorbs upload backlog (prevents cascading failures)
3. Read replica takes DB read load off primary
4. ElastiCache serves hot data (metadata, sessions) — plan in advance
5. CloudFront cache improves CDN hit rate (configure TTL proactively)

PRO TIP

S3 request limits

S3 has a default limit of 3,500 PUT and 5,500 GET requests per second **per prefix.** If your spike pushes 50x normal uploads, use multiple S3 key prefixes to distribute load. AWS returns 503 errors if you exceed per-prefix limits.

YEAR 4 · Security and Resilience · CTO

IAM security fundamentals

IAM Concept	What it means
Least Privilege	Every user and service gets only the minimum permissions it needs. No wildcard (*:*) policies.
Service Account	IAM role used by an application (Lambda, EC2). Should use programmatic access only — no console login.
Compromised Key Response	Immediately DEACTIVATE (not just rotate) the key, then audit CloudTrail for all API calls made with it.
MFA (Multi-Factor Auth)	Require MFA for all human console access. MFA Delete on S3 prevents data destruction even with stolen credentials.

Security monitoring tools

Tool	Purpose

AWS CloudTrail	Logs every API call — who, what, when, from where. Primary forensic tool for security incidents.
Amazon GuardDuty	ML-based threat detection. Analyses CloudTrail, VPC Flow Logs, DNS to flag suspicious activity.
VPC Flow Logs	Captures network traffic metadata (source/dest IP, port, protocol). Detects lateral movement.
AWS Config	Tracks resource configuration changes over time. Can auto-remediate non-compliant resources.

RULE

Compromised IAM key response

1. IMMEDIATELY deactivate/delete the compromised key (do not just rotate)
2. Review CloudTrail logs for all API calls made with that key
3. Assess blast radius — what did the attacker access or modify?
4. Rotate all other credentials as a precaution
5. Enable GuardDuty to detect future anomalies automatically

Resilience architecture

Pattern	Protection level
Multi-AZ RDS	Protects against single AZ failure. Automatic failover in 1-2 min. Standard for production.
Route 53 Failover	Detects unhealthy primary endpoint, redirects DNS to backup region automatically.
Global Accelerator	Routes traffic to healthy endpoints at network layer — faster failover than DNS.
S3 Versioning + MFA Delete	Prevents data destruction even if IAM credentials are compromised.
Multi-Region Active-Passive	One region handles traffic; second on standby. Activates on primary failure.

YEAR 5 · Global Architecture · CTO

Global service	When to use it
CloudFront CDN	Always for video delivery. 400+ edge locations. Reduces latency for all international users.
Route 53 (Global DNS)	Multi-region traffic routing, health checks, latency-based routing to nearest healthy region.

Aurora Global Database	Multi-region replication. Reads in each region, write to primary. Sub-second replication.
S3 Cross-Region Replication	Copies S3 objects to another region automatically. Supports GDPR data residency.
API Gateway	Manages 50,000+ concurrent API connections. Handles throttling, auth, and usage plans.

RULE

GDPR data residency

EU user personal data **must be stored and processed within the EU.** Deploy a dedicated stack in eu-west-1 (Ireland) or eu-central-1 (Frankfurt). Data must not replicate to us-east-1 for EU users. This is a **LEGAL requirement**, not a performance optimisation.

AWS Well-Architected Pillars

Pillar	Core question to ask
Operational Excellence	Can we deploy, monitor, and improve the system reliably?
Security	Are we protecting data, systems, and access correctly?
Reliability	Can the system recover from failures automatically?
Performance Efficiency	Are we using the right resources at the right size?
Cost Optimisation	Are we eliminating waste and using the most cost-effective options?

PART 04 · CFO ROLE

The Financial Analyst

You own the numbers. Track costs, model scenarios, assess financial risk. You answer: "Can we afford this?"

YEAR 1 · Core Financial Concepts · CFO

Metric	Formula	Example
Monthly Cash Burn	AWS Cost - Monthly Revenue	$18K - $12K = $6,000/month burn
Runway	Cash Reserves / Monthly Burn	$60K / $6K = 10 months runway
Cost-to-Revenue Ratio	(AWS Cost / Revenue) x 100	($18K / $12K) x 100 = 150% — losing money

Break-Even	When Cost = Revenue	Need to cut to $12K or grow revenue to $18K
Cost per User	Total AWS Cost / Active Users	$10K / 1,000 users = $10/user/month

RULE

Healthy cost-to-revenue ratios

Below 25%: Excellent infrastructure margin (common for scaled SaaS)
25-40%: Good, industry competitive
40-60%: Acceptable for growing startups
Above 80%: Unsustainable — fix this before raising money
150%: PulseStream Year 1 starting point — this is why Year 1 is 'The Cleanup'

FORMULA

RI cost calculations (most tested CFO calculation)

Hourly saving per instance = On-Demand price - RI price
Monthly saving per instance = Hourly saving x 720 hours
Total monthly saving = Monthly saving x Number of instances

Example: ($0.192 - $0.060) x 720 x 10 instances = **$950.40/month**

PRO TIP

RI break-even rule

A 1-year RI pays off when the instance runs more than ~55% of the time. 55% of 720 hours = ~396 hours/month. If an instance only runs 8 hours/day (business hours) = 240 hours = 33% — **On-Demand is cheaper.** Always check utilisation before committing.

YEAR 2 · Commitment Risk · CFO

Scenario	CFO assessment
RI utilisation at **80%+**	Good commitment. Getting the discount and using the capacity. Keep or expand.
RI utilisation at **45%**	Over-committed. Paying for idle capacity. On-Demand would be cheaper below 55%.
RI utilisation at **30%**	Significantly over-committed. Investigate right-sizing or list on RI Marketplace.
Customer base **drops 40%**	Stranded cost. Partial mitigation via RI Marketplace. Convertible RIs can be exchanged.

RULE

CFO budget structure

Fixed/Committed: RI costs, Support plan, EDP commits
Variable: On-Demand compute, data transfer, S3 storage, Lambda invocations
Contingency: 10-15% buffer for traffic spikes or unexpected incidents
Separate line items for each product feature (using Cost Allocation Tags)

YEAR 3 · Spike Economics · CFO

WORKED EXAMPLE

Calculating viral event ROI

Extra AWS cost = $22,000 (spike month overage)
New sign-ups = 50,000
If 20% convert to paying users = 10,000 new users

Monthly revenue per user = $2 → **$20,000/month additional revenue**
Payback period = $22,000 / $20,000 = just over 1 month
Verdict: worth it — LTV of retained users far exceeds the spike cost

FORMULA

Infrastructure gross margin

Gross Margin = (Revenue - AWS Cost) / Revenue x 100
Example: ($3.00 - $1.20) / $3.00 x 100 = **60%**
Healthy SaaS: 60-80%. Below 40%: architecture needs work or pricing needs to increase.

Data egress scales directly with video views. In a 10x spike, expect data transfer costs to spike ~10x.

YEAR 4 · Incident Costs and Compliance · CFO

When an outage or security breach happens, calculate the **FULL cost** — not just the AWS bill.

Cost category	Examples for PulseStream
Direct AWS cost	Additional data transfer during failover, DR environment costs
Lost revenue	Revenue per hour x hours of downtime
SLA penalty credits	Contractual credits paid to customers for breached uptime commitments
Incident response labour	Engineering hours x loaded hourly rate
Legal/compliance cost	Legal fees, GDPR notification process, potential regulatory fines
Churn (reputational)	Revenue lost from customers who cancel after the incident

RULE

SLA breach calculation

99.9% uptime = 43.8 min allowed downtime per month. A 4-hour outage **exceeds** the 99.9% SLA commitment. Customers are entitled to contractual credit. CFO should negotiate credits proportional to the obligation — not the customer's opening demand.

YEAR 5 · Scale-Up Financial Metrics · CFO

Metric	Definition and target
AWS Cost per Active User	Total monthly AWS cost / MAU. Target: below 20-25% of ARPU.
Cloud Gross Margin	(Revenue - AWS Cost) / Revenue. Target: 60-80% for mature SaaS.
Net Revenue Retention (NRR)	Revenue from existing customers this month / same cohort last month x 100. Target: 110%+.
Rule of 40	Annual Revenue Growth % + FCF Margin %. Score above 40 is healthy for SaaS.
LTV:CAC Ratio	Lifetime Value / Customer Acquisition Cost. Target: 3:1 or higher.

PRO TIP

Enterprise Discount Program (EDP)

An EDP is a minimum AWS spend commitment (e.g., $2M/year) in exchange for a discount (e.g., 15%). **ONLY sign if your projected spend clearly meets the committed level.** If you commit to $2M but spend $1.3M, you pay the $700K shortfall. Ask for a ramp-up clause: lower commitment Year 1, higher Year 2-3.

RULE

IPO financial control checklist

- Cost Allocation Tags on ALL resources
- Monthly cost review process documented
- Board-level cloud spend reporting with variance analysis
- Documented approval workflow for new AWS services
- CUR-based reporting — not just Cost Explorer screenshots

PART 05 · PM ROLE

The Growth Lead

You own the product roadmap, customer relationships, and growth strategy. You answer: "What do users need?"

YEAR 1 · Prioritisation Under Constraints · PM

When engineering bandwidth is limited (as it is in Year 1 cleanup mode), the PM must ruthlessly prioritise using the **MoSCoW method:**

Category	Meaning for PulseStream Year 1
Must Have	Features directly tied to revenue retention. If removed, customers leave immediately.
Should Have	Improve experience but not business-critical. Can wait 1-2 months.
Could Have	Nice-to-haves. Defer until Year 2 when finances stabilise.
Won't Have (now)	Explicitly deprioritised this year. Not 'never' — just 'not now.'

RULE

Feature vs Revenue rule

Usage % does not equal Revenue %. A feature used by 5% of users could generate **60% of revenue** (enterprise customers). Always segment by revenue impact, not just user count, before cutting anything.

Customer communication during cleanup

Situation	Communication approach
Routine cleanup, no impact	Brief update: 'We're making performance improvements. No action needed.'
Planned maintenance window	Advance notice with timing, expected duration, what users should do.
Unexpected slowness noticed	Acknowledge immediately. Say what you know. Commit to update cadence.
Security breach affecting GDPR.	Legal obligation to notify proactively. Work with legal. Within 72 hours for data

YEAR 2 · Product Strategy and Roadmaps · PM

Framework	Best used for	Key output
MoSCoW	Constraint-driven planning	Must/Should/Could/Won't list
Impact vs Effort Matrix	Choosing quick wins vs long-term investments	2x2 matrix: build quick-wins first
RICE Scoring	Large backlog with many comparable features	Numeric score: Reach x Impact x Confidence / Effort
Jobs-To-Be-Done	Understanding why customers buy the product	Core 'job' customer hires PulseStream to do

SLA basics for the PM

SLA concept	PM implication
99.9% uptime	43.8 min/month allowed downtime. Architecture must support this BEFORE you commit it contractually.
RIs and uptime	RIs are a billing model, NOT an uptime guarantee. Uptime comes from Multi-AZ, Auto Scaling, health checks.
Committing 99.95%	Only do this if engineering has confirmed the architecture can deliver it. Creates financial liability.

YEAR 3 · Crisis Communication · PM

The viral spike is simultaneously a **product crisis** and a **growth opportunity.** The PM manages both.

Time	What the PM does
T+0 (detected)	Confirm with engineering: is this real? How severe? Which users affected?
T+15-30 min	Publish first status: 'We are aware. Team investigating. Next update in 30 min.'
T+1 hour	Update with what you know. Never say 'AWS is down.' Give estimated resolution time.
T+resolution	Final update: issue resolved, what happened, what you're doing to prevent recurrence.
T+24-48 hours	Post-incident email to affected users. Brief, honest, forward-looking.

RULE

What NOT to say during an incident

■ 'AWS is down — not our fault' (shifts blame, unprofessional)
■ 'All systems operational' when they're not (destroys trust instantly)
■ Silence (silence = incompetence in customers' minds) ■ Over-promise resolution times you can't guarantee

PRO TIP

Post-viral activation playbook

1. Identify the activation event — what is the first 'aha moment'? (First video processed)
2. Segment 50,000 new users: Activated vs Dormant (registered but never used)
3. Run targeted re-engagement campaign for dormant users with specific CTA
4. Diagnose why 60% churned — session recordings + support tickets
5. Fix onboarding experience before the next spike, not after

YEAR 4 · Incident Ownership and Trust Recovery · PM

Audience	What they need
All users (status page)	What happened, how long it lasted, what you're doing to prevent recurrence.

Enterprise customers	Proactive outreach from PM/VP. Acknowledge SLA breach. Offer contractual credit.
Press/media	CEO or designated comms lead. Legal review. Factual, measured — no speculation.
Board/investors	Business impact summary: users affected, revenue lost, SLA liability, churn risk.
Regulatory body	Formal notification within 72 hours. What data involved, how many people, steps taken.

SOC 2 — why it matters

Certification	What it means
SOC 2 Type I	Point-in-time assessment. 'Your security controls look good today.'
SOC 2 Type II	Sustained assessment over 6-12 months. Enterprise standard.
ISO 27001	International information security management standard. Pairs well with SOC 2.
GDPR Compliance	Legal requirement for EU user data. Not optional.
PCI-DSS	Required if PulseStream processes payment card data.

YEAR 5 · Growth Strategy and IPO Readiness · PM

Net Revenue Retention (NRR) — the most important PM metric

NRR Score	What it means
Below 80%	Customers churning faster than you retain. Product-market fit problem. Fix first.
80-100%	Retaining most revenue but not growing existing accounts. Stable but not great.
100-110%	Retaining AND some customers expanding. Healthy SaaS business.
110%+	Expansion revenue exceeds churn. The compounding flywheel. IPO-ready metric.

GDPR product requirements (EU launch)

GDPR requirement	Product change required
Lawful basis	Consent mechanism on sign-up — clear, specific, not pre-ticked boxes.
Right to Erasure	'Delete my account' must delete ALL user data within 30 days.
Data Portability	Users can request a download of all their data. Build data export feature.

Breach Notification	Notify authority within 72 hours of discovering a breach affecting personal data.
Data Residency	EU user data stays in EU region (eu-west-1 or eu-central-1). Not replicated to US.

RULE

IPO product readiness milestones

- NRR consistently above 110%
- SOC 2 Type II certification complete
- GDPR and regional compliance documented
- Public API with documented SLAs
- Product roadmap aligned to a defensible market category
- Technical debt and critical reliability issues resolved

Competitive response framework

Situation	PM response
Competitor launches similar feature	Analyse depth/quality first. Fast-follow only if customer demand signals it's important.
Competitor undercuts price by 30%	Don't automatically match. Understand if their pricing is sustainable. Reinforce your differentiators.
Hyperscaler offers customer 40% discount	Price match for key accounts buys time but doesn't solve the real problem.
Competitor gains market share fast	Focus on your defensible niche. Harder to be best at a specific segment than compete broadly.

PART 06

Quick Reference

Formulas, decision trees, and common traps. Keep this page open during the game.

The formulas

Formula	Calculation	When you need it
Monthly Cash Burn	AWS Cost - Revenue	Year 1 — find the gap
Runway (months)	Cash / Monthly Burn	Year 1 — how long to fix this
RI Monthly Saving	(On-Demand - RI price) x 720 x # instances	Year 2 — evaluate RI commitment

RI Stranded Cost	Unused RI x RI price x 720 x months remaining	Year 2 — risk of over-commitment
Infra Gross Margin	(Revenue - AWS Cost) / Revenue x 100	Years 3-5 — investor metric
Cost per Active User	Total AWS Cost / MAU	Years 3-5 — unit economics
LTV of viral users	(Conv rate x New users) x ARPU x avg lifetime	Year 3 — viral event ROI
Rule of 40	Revenue growth % + FCF margin %	Year 5 — IPO health metric

Uptime SLA reference

SLA	Max monthly downtime	Max annual downtime
99%	7.3 hours/month	3.65 days/year
99.9%	43.8 minutes/month	8.77 hours/year
99.95%	21.9 minutes/month	4.38 hours/year
99.99%	4.4 minutes/month	52.6 minutes/year

A 4-hour outage violates 99.9% and 99.95% SLAs. It does NOT violate a 99% SLA.

AWS service decision tree

If you need to...	Use this
Run a virtual server	**EC2** (choose the right family and purchase model)
Store files, videos, backups	**S3** (Standard for hot data, Intelligent-Tiering for mixed access)
Run a SQL database	**RDS** (with Multi-AZ for production)
Cache frequently accessed data	**ElastiCache** (Redis)
Queue jobs to absorb traffic spikes	**Amazon SQS**
Deliver content globally	**CloudFront CDN**
Auto-scale compute based on load	**Auto Scaling Group + CloudWatch alarm**
Route traffic to healthy regions	**Route 53 with failover routing**
Detect security threats	**Amazon GuardDuty**

Audit all API calls	**AWS CloudTrail**
Analyse cost spend	**AWS Cost Explorer**
Set spend alerts	**AWS Budgets**
Get right-sizing recommendations	**AWS Compute Optimizer**
Manage API connections at scale	**Amazon API Gateway**

Rate-the-Choice decision framework

Rate-the-Choice questions ask you to evaluate whether a decision is Good, Risky, or Bad:

Rating	When to choose it
Excellent / Great	Clearly the right call. Benefits strong, risks minimal or well-managed.
Good (with a caveat)	Correct direction, but there's a condition that must hold. Usually the right answer for 'if X holds.'
Risky	Could work but has a meaningful downside not being acknowledged.
Bad	Wrong in direction. Usually a fundamental misunderstanding of the trade-off.

PRO TIP

Rate-the-Choice thinking pattern

Ask: What is the stated goal? Does this decision achieve it?
Ask: What is the main risk? Is it acknowledged?
Ask: Is the timing right? The right action at the wrong time is 'Risky.' **Ask:** Who does this decision hurt? Customers, engineers, finance?

The answer is usually not 'Bad' unless there's a fundamental error. Most realistic decisions are 'Good with a caveat' or 'Risky.'

Common traps in the game

Trap	The right thinking
Buying **3-year RIs for a startup**	Startups change too fast. 1-year max unless very stable.
Auto-stopping instances when budget exceeded	This kills the product during its most important moments. Alerts are not auto-shutdown.
Rebuilding architecture during financial crisis	Fix current waste first. Re-architecting during Year 1 adds risk at the worst time.
Matching competitor's 30% price cut immediately	Understand their cost structure first. Reactive cuts destroy margins permanently.

Communicating nothing during incident	Silence destroys trust faster than the incident itself. Acknowledge early, update regularly.
Assuming **RIs guarantee uptime**	RIs are a billing tool. Uptime requires Multi-AZ, Auto Scaling, health checks.
Treating **all 5% users** as low-value	5% of users might be 60% of revenue. Segment by value, not volume.
Committing **EDP before expansion**	If you sign $2M/yr EDP then shift compute to a new region, your usage changes. Model first.

PART 07

Glossary

Every term used in this handbook, in plain English.

Term	Plain English definition
ARR	Total predictable revenue from subscriptions per year.
ARPU	Monthly or annual revenue divided by number of active users.
Auto Scaling	AWS feature that automatically adds/removes EC2 instances based on demand.
AZ (Availability Zone)	A physical data centre within an AWS region. Each region has 2-6 AZs.
Cash Burn	Monthly spending minus monthly revenue.
CUR	The most detailed AWS billing file — every resource, every hour.
EDP	AWS volume discount in exchange for a minimum annual spend commitment.
EIP (Elastic IP)	A static public IP address. Costs money even when not attached to anything.
ElastiCache	AWS managed in-memory caching service (Redis or Memcached). Reduces DB load.
FCF Margin	Cash remaining after all operating expenses. Key IPO metric.
GDPR	EU data privacy law. Requires consent, right to erasure, breach notification, data residency.
GuardDuty	AWS threat detection using ML to find suspicious API and network activity.

IAM	AWS service controlling who and what can do what in your account.
LTV	Total revenue expected from one customer over their entire relationship.
MFA	Security requiring a second verification step beyond password.
Multi-AZ	Deploying resources across multiple Availability Zones for fault tolerance.
NRR	Revenue from existing customers this period vs last period. Above 110% = expansion.
On-Demand	AWS billing model with no commitment. Pay per second, cancel anytime.
Post-Mortem	Blameless analysis of what went wrong and what systemic fix prevents recurrence.
Reserved Instance (RI)	AWS billing commitment (1 or 3 years) for 40-60% discount vs On-Demand.
RPO	Maximum acceptable data loss in a failure (e.g., 15-min RPO).
RTO	Maximum acceptable downtime before recovery (e.g., 1-hour RTO).
Runway	How many months until the company runs out of cash at current burn rate.
S3 Intelligent-Tiering	S3 class that auto-moves objects to cheaper tiers when access drops.
Savings Plan	AWS flexible commitment discount across EC2, Lambda, and Fargate.
SOC 2 Type II	Security audit proving controls operated effectively over time. Enterprise standard.
Spot Instance	EC2 capacity at 70-90% discount. Can be reclaimed with 2 minutes notice.
SQS	AWS message queue. Decouples producers/consumers. Absorbs traffic spikes gracefully.
Stranded Cost	RI or EDP spend that can't be utilised because demand dropped.
Trusted Advisor	AWS tool providing automated recommendations across cost, security, performance.
VPC Flow Logs	Network traffic metadata logs for security analysis and anomaly detection.

Good luck, PulseStream leadership team. Think like an executive.

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